

Fact Strategies: Turn It Around, Break It Apart

turn-around facts, break-apart strategies, and choosing between them

Directions

- **Turn-around facts:** the order of the factors never changes the product.
- **Break-apart facts:** split one factor into easy pieces, multiply each piece, and add the parts.
- Problems 3 and 4 show the pattern worked out — after that, the strategy is yours to run.

1. Kim knows $3 \times 7 = 21$. Use the turn-around fact: $7 \times 3 =$ _____

2. You know $6 \times 4 = 24$. What is 4×6 ? _____

3. Finish the worked pattern: $7 \times 6 = 7 \times 5 + 7 \times 1 = 35 + 7 =$ _____

4. Break the 7 apart, then fill in all three blanks:

$$8 \times 7 = 8 \times 5 + 8 \times 2 = \underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$$

5. Write the turn-around fact for 9×2 :

Answer: _____

Both facts have the same product: _____

6. Finish the break-apart with a take-away: $6 \times 9 = 6 \times 10 - 6 \times 1 =$ _____

7. Trina wants to skip count to find the 8-and-5 fact. Circle the order that is easier to skip count:

8×5 (count by 5s, eight times) 5×8 (count by 8s, five times)

Then find the product: _____

8. Break the 12 apart to solve: $4 \times 12 = 4 \times 10 + 4 \times 2 =$ _____

9. Solve 6×7 two different ways.

(a) Break the 7 into $5 + 2$: $6 \times 7 =$ _____

(b) Break the 6 into $5 + 1$: $7 \times 6 =$ _____

(c) Say why both ways must give the same product.

10. Challenge. Use any strategy from this sheet — turn-around or break-apart — to find 9×7 .

Show your strategy:

Answer: _____

$9 \times 7 =$ _____